Honghao Zhao

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Education

Johns Hopkins University Aug 2024 - May 2026
Master of Science in Engineering in Computer Science GPA: 3.75/4.00

Boston University Sep 2021 - Jun 2024
B.S., Double Major: Pure and Applied Mathematics and Computer Science GPA: 3.70/4.00

Research Interests: Human-Robot Interaction | Robot Learning | Surgical Robotics | Safe Robot Autonomy | VR/AR

Publication list

**Equal contribution. †Equal second authorship.*

C. Liu*, C. Yu*, **H. Zhao***, H. Ding, H. Wei, A. Munawar, M. Unberath, and P. Kazanzides, "Digital Twin-Driven VR Teleoperation with Multi-View Spatial Perception for Surgical Robots," ***IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)***, 2026. ***(Accepted)*** arXiv.

A. Mahmood*, B. Kim*, **H. Zhao**, M. E. Atwood, L. F. Buenaver, M. T. Smith, and C.-M. Huang, "Better Adherence, Richer Context: A Field Evaluation of LLM-Powered Conversational Voice Diaries for Sleep," ***ACM Transactions on Computing for Healthcare (HEALTH)***, 2026. ***(Major Revision)*** arXiv.

B. Kim*, A. Mahmood*, **H. Zhao**, M. E. Atwood, L. F. Buenaver, Z. Xiao, and C.-M. Huang, "Guided Reflection for Personal Sleep Insight in Everyday Sleep Tracking," ***ACM Conference on Human Factors in Computing Systems (CHI)***, 2027. ***(Under Review)*** arXiv.

A. Mahmood, B. Kim, **H. Zhao**, M. E. Atwood, L. F. Buenaver, M. T. Smith, and C.-M. Huang, "Conversational Agents in Behavioral Sleep Medicine: Designing Self-Report and Analytics Tools," ***ACM Conference on Human Factors in Computing Systems (CHI)***, 2027. ***(Under Review)*** arXiv.

Manuscripts in preparation *(working titles)*

H. Zhao, V. Kamath†, S. Terala†, and U. Tumerdem, "Autonomous Robotic Surgical Manipulation with Haptics-Enhanced Imitation Learning." *In preparation for submission to **IEEE Robotics and Automation Letters (RA-L)**.*

H. Zhao*, N. Baek*, S. Hu, Y. Wang, B. Qu, and H. Hu, "Multimodal Human-Centered Safety Filtering with Vision-Language Models and Haptic Feedback." *In preparation for submission to **Robotics: Science and Systems (RSS)**.*

Research Experience

Alliance AI Lab, Johns Hopkins University Jun 2026 - Present
*Research Assistant | Advisor: **Prof. Haimin Hu***
Multimodal Human-Robot Interaction and Safety
- Extend the Human-Centered Safety Filter (HCSF) with fine-tuned VLMs and haptic, visual, and voice feedback.

Learning for Dynamics and Control in Autonomous Racing
- Collaborate with a Princeton group on learning and control research using the F1TENTH platform.

Laboratory for Computational Sensing and Robotics (LCSR), Johns Hopkins University Oct 2025 - Present
*Research Assistant | Advisor: **Prof. Peter Kazanzides***
Digital-Twin VR Teleoperation for Surgical Robots
- Built dVRK teleoperation with ROS-Unity integration, multi-view rendering, and markerless object tracking.
- Evaluated motion efficiency and smoothness in a user study against a mixed-reality teleoperation baseline.

Haptics-Enhanced Surgical Imitation Learning
- Led AMBF data collection and force-augmented ACT training; tested generalization to unseen positions.

Intuitive Computing Laboratory, Johns Hopkins University Apr 2025 - Sep 2026
*Research Assistant | Advisor: **Prof. Chien-Ming Huang***
Conversational AI and Clinical Analytics for Sleep Health
- Built sleep-diary data pipelines and clinician-facing visualizations.
- Contributed to DREAM (Discovery, Reflection and Education for Awareness and Management of Sleep), a voice assistant evaluated in a longitudinal field study.

Work Experience

NeuroAgent AI | Baltimore, Maryland, USA May 2026 - Present
*Research Intern | Advisor: **Dr. Kemar Green***
Eye-Movement Modeling and Tracking for Digital Biomarkers
- Built an eye-movement simulator and evaluated recurrent models against a geometric baseline on unseen participants.
- Developed video-based eye tracking with head-pose correction; explore data collection in Zambia.

Teaching Experience

Boston University, Department of Computer Science Jan 2023 - Jul 2023
Teaching Assistant, CS460: Introduction to Database Systems

Technical Skills

Programming: Python, C/C++, Java, SQL. **Software & Tools:** ROS, Unity, AMBF, SAM 2, FoundationPose.
Frameworks & Methods: PyTorch, RL, imitation learning (ACT), LLMs, VLM fine-tuning, MPPI, QCBF.